

CLICK TO RESET

TIME'S UP!

1 Q.) Two pipes A and B can fill a tank in 36 hours and 45 hours respectively. If both the pipes are opened simultaneously, how much time will be taken to fill the tank?

A. 50hrs

B. 6hrs

C. 20hrs

D. 10hrs

|   |    |                 |       |
|---|----|-----------------|-------|
| A | 36 | $\frac{180}{9}$ | 5 ✓   |
| B | 45 |                 | 4     |
|   |    |                 | <hr/> |
|   |    |                 | 9     |
|   |    |                 | <hr/> |
|   |    | $\frac{180}{9}$ | 20    |

Diagram: A tank with two pipes. Pipe A is labeled 5 and Pipe B is labeled 4.

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2 Q.A tub can be filled in 20 minutes but there is a leakage in it which can empty the full tub in 60 minutes. In how many minutes it can be filled?

a) 10 minutes b) ✓ 30 minutes c) 40 minutes d) 25 minutes e) None of these

$$\begin{array}{r} A \\ - B \\ \hline \end{array}$$
$$\frac{60}{2} = 30$$

Handwritten calculations and diagrams illustrating the solution:

- A subtraction problem:  $A - B$
- A division problem:  $\frac{60}{2} = 30$
- A diagram showing a vertical line with a horizontal line extending from the bottom, and a curved line above it, possibly representing a tub or a path.
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3 Q.) Pipe A can fill a tank in 20 hours while Pipe B can fill it in 30 hours and Pipe C can empty the same tank in 40 hours. If all the pipes are opened together, what will be the time needed to fill the tank completely?

A  
B  
- C

120

6

4

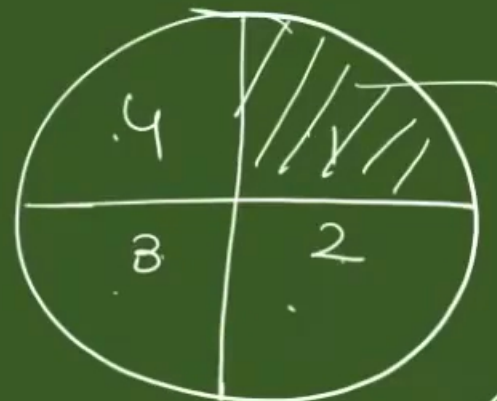
3

$$\frac{120}{7} = 17 \frac{1}{7}$$

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## Pizza Concept :



25%  $\rightarrow$  4 day

$$\begin{array}{r} 4 \\ \times 100 \\ \hline 400 \end{array}$$

16.1

$$\frac{\text{days}}{\text{fraction}} \times 100$$



4Q . P can complete  $\frac{1}{4}$  of a work in 10 days, Q can complete 40% of the same work in 15 days, R;  $\frac{1}{3}$  of the work in 13 days and S,  $\frac{1}{6}$  of the work in 7 days. Who will be able to complete the work first?

- A. P
- B. Q
- C. R
- D. S

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P 40  
Q  $75/2$   
R

$$\frac{10}{\frac{1}{4}} \times 1$$

$$\frac{15}{\frac{40}{100}} \times \frac{5}{100}$$



5 Q . P can complete  $\frac{1}{4}$  of a work in 10 days, Q can complete 40% of the same work in 15 days, R,  $\frac{1}{3}$  of the work in 13 days and S,  $\frac{1}{6}$  of the work in 7 days. Who will be able to complete the work first?

- A. P
- B. Q ✓
- C. R
- D. S

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|     |                |
|-----|----------------|
| ✓ P | 40             |
| ✓ Q | $\frac{75}{2}$ |
| ✓ R | 39             |
| ✓ S | 42             |

$$\frac{13}{\frac{4}{3}} \times 1$$
$$\frac{7}{\frac{4}{6}}$$



6

Q . P can complete  $\frac{1}{4}$  of a work in 10 days, Q can complete 40% of the same work in 15 days, R,  $\frac{1}{3}$  of the work in 13 days and S,  $\frac{1}{6}$  of the work in 7 days. Who will be able to complete the work first?

- A. P
- B. Q
- C. R
- D. S

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|     |                |
|-----|----------------|
| ✓ P | 40             |
| ✓ Q | $\frac{75}{2}$ |
| ✓ R | 39             |
| ✓ S | 42             |

$$\frac{13}{\frac{4}{3}} \times 1$$

$$\frac{7}{\frac{4}{6}}$$



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7 Q. A can do  $\frac{1}{3}$  of the work in 5 days and B can do  $\frac{2}{5}$  of the work in 10 days. In how many days both A and B together can do the work?

A)  $7 \frac{3}{4}$

B)  $8 \frac{4}{5}$

✓ C)  $9 \frac{3}{8}$

D) 10

E) 12

$$\begin{array}{r} A - 15 \\ B - 25 \end{array} \rightarrow 75 \quad \begin{array}{r} 5 \\ 3 \end{array} \Bigg|$$

$$\frac{75}{8} = 9 \frac{3}{8}$$



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- 8 Q. A can do  $\frac{1}{2}$  of a piece of work in 5 days, B can do  $\frac{3}{5}$  of the same work in 9 days and C can do  $\frac{2}{3}$  of that work in 8 days. In how many days can three of them together do the work ?

A. 3

B. 5

C.  $4\frac{1}{2}$

D. 4



Handwritten calculations for the problem:

|   |      |           |   |
|---|------|-----------|---|
| A | — 10 | 6         | ✓ |
| B | — 15 | 4         | ✓ |
| C | — 12 | 5         | ✓ |
|   |      | <u>15</u> |   |

A curved arrow points from the sum 15 to the number 60, which is written next to the list of days. An arrow also points from the number 60 to the middle section of the rectangle diagram.

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**With Help Of :**



9 Q.) A man can do a piece of work in 5 days, but with the help of his son, he can do it in 3 days. In what time can the son do it alone?

A.  $16/3$  B. 6 C. 9 D.  $15/2$

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$$\begin{array}{r} A \quad \text{---} \\ A + \text{Son} \quad \text{---} \\ \hline 3 \quad \quad 2 \end{array}$$

$$\begin{array}{r} 15 \\ \hline 3 \\ 5 \end{array}$$

$$\frac{15}{2} = 7 \frac{1}{2}$$





10 Q.) A can lay railway track between two given stations in 16 days and B can do the same job in 12 days, with help of C, they did the job in 4 days only. Then, C alone can do the job in:

$$\begin{array}{r} 48 \\ 5 \end{array}$$

A —

B —

A+B+C —

1  
3

1  
4

5

3  
4  
12

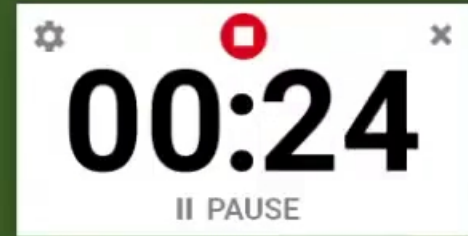


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**11 Q.) A can do a piece of work in 10 days, and B can do the same work in 20 days. With the help of C, they finished the work in 4 days. C can do the work in how many days, working alone?**

- 1. 5 days**
- 2. 10 days**
- 3. 15 days**
- 4. 20 days**
- 5. 25 days**



## Efficiency:

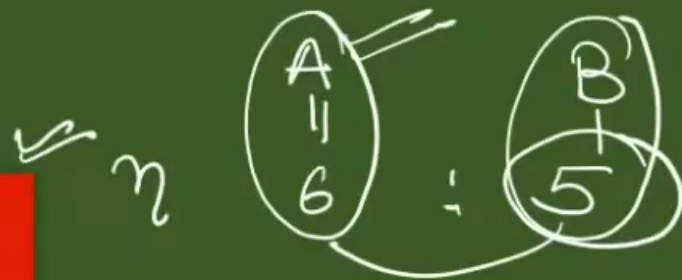
Ques (A is 20% more eff than B. if A can do — 12

$$\frac{6 \times 12}{5}$$

a) B alone

b) A+B = ?

$$\frac{6 \times 12}{11}$$



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Efficiency: <sup>12</sup> Q) A is 30% more efficient than B. How much time will they, working together, take to complete a job which A alone could have done in 23 days?

a) 11 days b) ~~13~~ days c) 20 d) Data inadequate e) None of these

<sup>13</sup> Q) Sakshi can do a piece of work in 20 days. Tanya is 25% more efficient than Sakshi. The number of days taken by Tanya to do the same piece of work is:

a) 15 b) 16 c) 18 d) 25 e) None of these

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Handwritten calculations:

Efficiency:  $\frac{A}{13}$  (circled),  $B \checkmark$   $\frac{10}{10}$

$\frac{30}{100} = \frac{3}{10}$  (circled)

Work =  $\frac{13 \times \cancel{23}}{\cancel{23}}$

13 day

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14 Q) A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in:

a) 4 days b) 6 days c) 8 days d) 12 days e) None of these

|     |   |               |               |
|-----|---|---------------|---------------|
|     | A | <u>B</u>      | C             |
| day | 1 | $\frac{1}{2}$ | $\frac{1}{3}$ |
| n   | 1 | <u>2</u>      | 3             |

$$\text{work} = 6 \times 2 = \frac{12}{2} = 6 \text{ days}$$





Challenge:

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15 Q.) Time taken by P alone to finish a work is 50% more than the time taken by P and Q together. Q is thrice as efficient as R. If Q and R together can complete the work in 22.5 days, then how many days will P alone take to complete the work?

A.17  
B.11

C.15  
D.NOT

Time

|            | P | P + Q | Q | R |
|------------|---|-------|---|---|
| Efficiency | 2 | 2     | 3 | 1 |

$(2 + 1) \times 3$

P : Q : R

6 : 3 : 1

$4 \times 22.5 = 90$

$90 \div 6 = 15$  days

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Pipe A is three times faster than Pipe B and takes 24 minutes less time than Pipe B to fill the cistern. If both pipes are opened together, in how much time will the cistern be completely filled?

~~16~~ 16

Q.) One fill pipe A is 3 times faster than second fill pipe B and takes 24 minutes less than the fill pipe B. When will the cistern be full if both pipes are opened together?

✓  $\eta$  ✓ day

|   | A      | B   |
|---|--------|-----|
| ✓ | 3      | 1   |
| ✓ | (1)    | (3) |
|   | A — 12 | 36  |
|   | B — 36 | 1   |

2 → 24  
1 → 12

$\frac{36}{4} = 9$

days





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17 Q.) One fill pipe A is 3 times faster than second fill pipe B and takes 16 hours less than the fill pipe B. When will the cistern be full if both pipes are opened together?

A. 5hrs

B. 6hrs

C. 4hrs

D. Data Inadequate

Time





$$\checkmark 4 + 3 \checkmark = ?$$

18 Q) A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is:

a)  $\frac{1}{4}$  b)  $\frac{1}{10}$  c)  $\frac{7}{15}$  d)  $\frac{8}{15}$  e) None of these

$$\begin{array}{rcl} A & - & 15 \\ B & - & 20 \end{array} \quad \begin{array}{c} \checkmark \\ \textcircled{60} \\ \uparrow \end{array} \quad \begin{array}{c} 4 \\ 3 \end{array}$$

$$7 \times 4 = \textcircled{28}$$

$$R = \frac{\cancel{32}}{\cancel{60}} = \frac{8}{15}$$

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90

19 Q.) A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days, A alone can finish the remaining work?

a) 5 b) 5 c) 6 d) 8 e) None of these

|     |    |    |              |
|-----|----|----|--------------|
| = A | 18 | 90 | 5            |
| = B | 15 |    | (6) $\alpha$ |

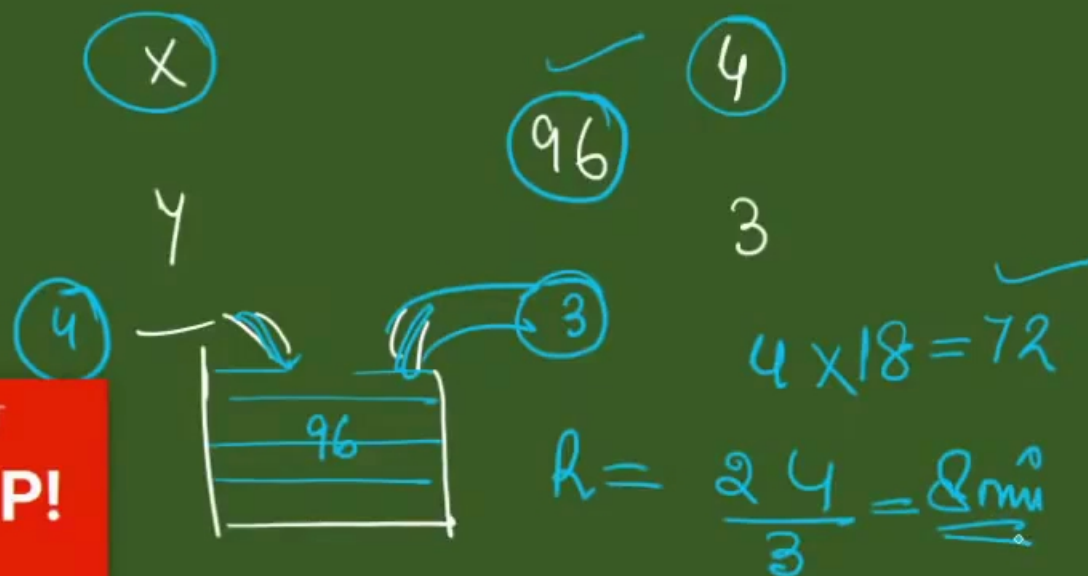
$$6 \times 10 = 60$$

$$R = \frac{90 - 60}{5} = 6$$



20 Q.) Two pipes X and Y can fill a cistern in 24 minutes and 32 minutes respectively. If both the pipes are opened together, then after how much time (in minutes) should Y be closed so that the tank is full in 18 minutes?

- (a) 10      (b) 8      (c) 6      (d) 5      (e) none of these



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21 Two pipes A and B can fill a cistern in  $37\frac{1}{2}$  minutes and 45 minutes respectively. Both pipes are opened. The cistern will be filled in just half an hour, if the B is turned off after:

- A. 5 min.
- B. 9 min.
- C. 10 min.
- D. 15 min.

HW





22

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Q) A and B can complete a work in 15 days and 10 days respectively. They started doing the work together but after 2 days B had to leave and A alone completed the remaining work. The whole work was completed in:

a) 8 days b) 10 days c) ~~12 days~~ d) 15 days e) None of these



|   |    |
|---|----|
| A | 15 |
| B | 10 |

|      |     |
|------|-----|
| (30) | (2) |
|      | 3   |

$$5 \times 2 \text{ day} = \underline{\underline{10}}$$

$$R = \frac{20}{2} = \underline{\underline{10 \text{ days}}}$$

$$\text{Total} = 10 \text{ day} + 2 \text{ day} = 12$$





23

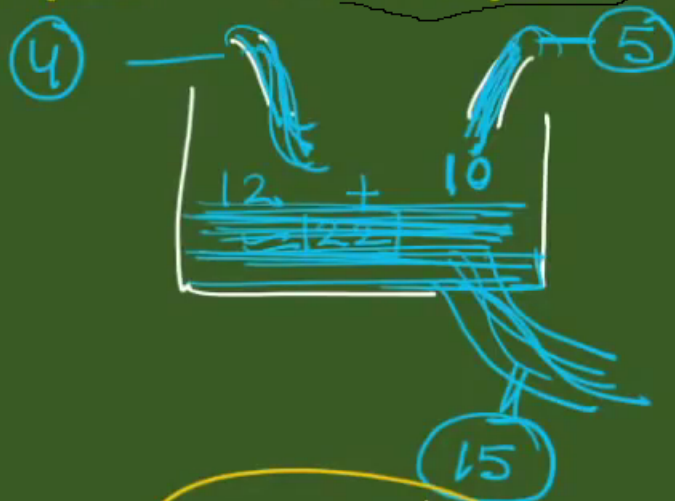
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Q. Two pipes can fill a tank with water in 15 and 12 hours respectively and a third pipe can empty it in 4 hours. If the pipes be opened in order at 8, 9 and 11 a.m. respectively, the tank will be emptied at

- (a) 11 : 40 a.m. (b) 12 : 40 p.m.  
(c) 1 : 40 p.m. (d) 2 : 40 p.m.

(e) none of these



2 : 40 p.m.

$$\begin{array}{r}
 15 \\
 12 \\
 4 \\
 \hline
 22
 \end{array}
 \begin{array}{r}
 60 \\
 4 \\
 5 \\
 15
 \end{array}$$

$$\frac{22}{6} = 3 \frac{4}{6}$$

$$3 \text{ hr} + \frac{4}{6} \times 60 \text{ min}$$

@ 11

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$$\checkmark \left| \frac{m_1 D_1 H_1}{w_1} = \frac{m_2 D_2 H_2}{w_2} \right|$$

## MDH RULE BASED QUESTIONS

$$\frac{100 \times 100}{100} = \frac{4 \times 100}{4}$$



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~~24~~ Q.) 5 men can prepare 10 toys in 6 days working 6 hours a day. then in how many days can 12 men prepare 16 toys working 8 hrs a day ?

A. 15

B. 12

C. 10

D. 6

E. 3

$$\frac{5 \times 6 \times 6}{10} = \frac{12 \times 8 \times \text{Men}}{16}$$



25 Q.) Bunnu, Gaffur and Ram can do a work in 10, 12 and 15 days respectively knowing that they work together and get an amount of Rs. 1500. What is the share of Gaffur in that amount?

- (a) Rs 300    ☒ (b) Rs 500    (c) Rs 450    (d) Rs 400    (e) None of these

$$\begin{array}{lcl} B & - & 10 \\ \textcircled{G} & - & 12 \\ R & - & 15 \end{array} \quad \begin{array}{l} 6 \\ 60 \text{ } \textcircled{5} \\ 4 \end{array}$$

$$\begin{array}{r} 5 \times 1500 \\ \hline 15 \end{array} \quad 6 : 5 : 4$$



26 Q) A alone can do a piece of work in 6 days and B alone 8 days. A and B undertook to do it for Rs.3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?

a) Rs.375 b) Rs.400 c) Rs.600 d) Rs.800 e) None of these

CAT

~~Imp.~~

|       |   |   |
|-------|---|---|
| A     | 6 | 4 |
| B     | 8 | 3 |
| A+B+C | 3 | 8 |

|   |   |   |                           |
|---|---|---|---------------------------|
| A | B | C | $\frac{1}{8} \times 3200$ |
| 4 | 3 | 1 |                           |

400





27 If a monkey climb a pole of height 50m, it climbs up 5m in 1st minute and slips down 4m in 2nd minute. In how many minutes monkey will climb the pole? 91 minutes



$$\begin{array}{r} 1 \text{ metre} - 2 \text{ minutes} \\ \times 45 \quad \times 45 \\ \hline 45 \text{ metres} - 90 \text{ minutes} \end{array}$$



**28** If a monkey climb a pole of height 100m, it climbs up 6m in 1st minute and slips down 4m in 2nd minute. In how many minutes monkey will climb the pole?

**95 minutes**



29 A cistern is provided with two pipes A and B. A can fill it in 20 minutes, B can empty it in 30 minutes. If A and B kept alternatively for one minute each, how soon the cistern will be filled? 115 minute



57m



1 metre - 2 minutes

X 57

X 57

---

57 metres - 114 minutes

30 Two pipes A and B can fill a tank in 20 and 30 minutes respectively.  
If both the pipes are open together, the time taken to fill the tank is?

|        |    |                  |
|--------|----|------------------|
| A - 20 | 3  |                  |
|        | 60 | ← Total Capacity |
| B - 30 | 2  |                  |

$$\frac{60}{5} = 12$$

31 Pipe A and B can fill a tank in 5 hrs and 6 hrs. Pipe C can empty it in 12 hrs.  
If all the three pipes are opened together, the tank will be filled in?

|        |    |    |
|--------|----|----|
| A - 5  |    | 12 |
| B - 6  | 60 | 10 |
| C - 12 |    | -5 |

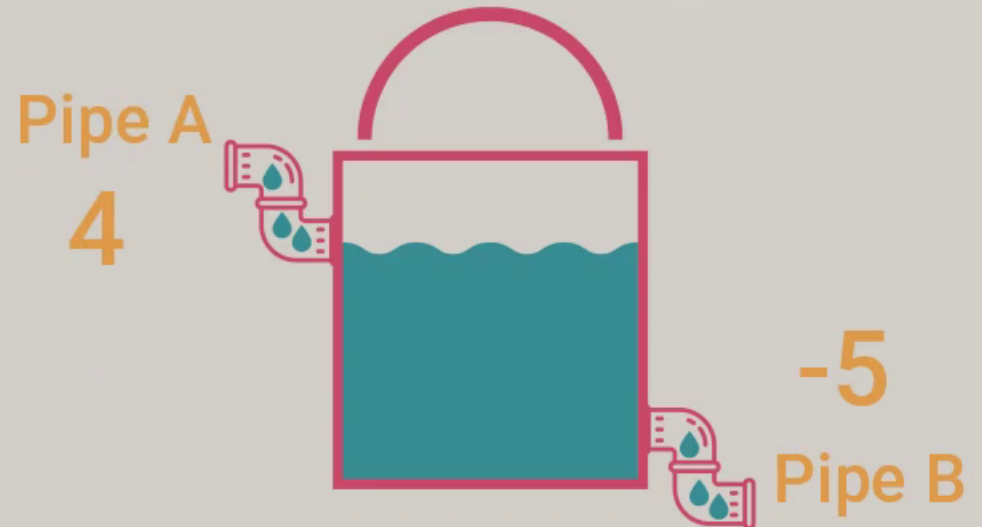
$$\frac{60}{17} = 3\frac{9}{17}$$





32 A cistern can be filled with water by a pipe in 5 hrs and can be emptied by a second pipe in 4 hrs. If both the pipes are opened when the cistern is already full, the time in how much time cistern will be empty? 20 hours

|       |    |
|-------|----|
| A - 5 | 4  |
| 20    |    |
| B - 4 | -5 |



33 A cistern is normally filled in 9 hrs. But due to the leakage in the bottom, it takes 1 hr more to be filled, in what time the cistern will get empty?

$$A - 9 \quad 10$$

$$90$$

$$A + \text{Leakage} - 10 \quad 9$$

$$\frac{90}{1} = 90 \text{ minutes}$$





34 A pipe can fill a tank with water in 3 hrs. Due to leakage in the bottom, it takes  $3\frac{1}{2}$  hrs to fill it. In what time the leak will empty the fully filled tank? 21 hours

$$A - 3/1 \quad 7$$

$$21/1$$

$$A + B - 7/2 \quad 6$$



Pipe A

